CET-2

1 Explain Topology Structures.

Topology Structures

A **topology structure** refers to the physical or logical arrangement of computers, devices, and communication links in a network. It defines how different nodes are connected and how data flows between them.

Types of Topology Structures

1. **Bus Topology**
    - All devices are connected to a single main cable called the bus.
    - Data travels in both directions through the bus.
    - Simple and inexpensive to install.

Advantages

- Easy to set up
- Requires less cable

Disadvantages

- Failure of the main cable can stop the entire network
- Difficult to troubleshoot

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2. **Star Topology**
    - All devices are connected to a central hub or switch.
    - Data passes through the central device.

Advantages

- Easy to manage and troubleshoot
- Failure of one device does not affect others

Disadvantages

- Failure of the hub or switch stops the whole network
- Requires more cable

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3. **Ring Topology**
    - Each device is connected to two other devices, forming a circular path.
    - Data travels in one direction around the ring.

Advantages

- Data transmission is orderly
- Less chance of data collision

Disadvantages

- Failure of one node can affect the whole network
- Difficult to add or remove devices

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4. **Mesh Topology**
    - Every device is connected to every other device.
    - Provides multiple paths for data transfer.

Advantages

- Very reliable
- Failure of one connection does not affect communication

Disadvantages

- Expensive
- Requires a large amount of cabling

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5. **Tree Topology**
    - Combination of star and bus topology.
    - Devices are connected in a hierarchical structure.

Advantages

- Easy to expand
- Suitable for large networks

Disadvantages

- Failure of the main cable affects the network
- Complex structure

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6. **Hybrid Topology**
    - Combination of two or more topologies.
    - Used in large organizations.

Advantages

- Flexible and scalable
- Reliable

Disadvantages

- Expensive
- Difficult to design and maintain

Q- 2 Differentiate between inducing a calculating response and eliciting an emotional response in data visualizations.

Difference between Inducing a Calculating Response and Eliciting an Emotional Response in Data Visualizations

Basis	Inducing a Calculating Response	Eliciting an Emotional Response
Meaning	Encourages the viewer to think logically and analyze the data carefully	Aims to create feelings, emotions, or personal connection with the data
Focus	Facts, numbers, comparisons, and accuracy	Human impact, storytelling, and emotional appeal
Purpose	Help users make rational decisions based on data	Make the audience feel concerned, inspired, shocked, or motivated
Type of Visualization	Tables, bar charts, line graphs, scatter plots	Images, colors, infographics, pictograms, storytelling visuals
Viewer Reaction	“What does the data mean?”	“How does this data make me feel?”
Usage	Business reports, financial analysis, scientific studies	Awareness campaigns, social issues, healthcare, environment

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Inducing a Calculating Response

This type of visualization helps people analyze and compare data using logic and reasoning. It focuses on exact values, percentages, trends, and calculations.

Example:
A line graph showing yearly sales growth of a company helps managers make business decisions.

Eliciting an Emotional Response

This type of visualization is designed to make people emotionally connected to the data. It often uses colors, icons, images, and storytelling to create impact.

Example:
A chart showing the number of deaths due to accidents using human figures or red color can create sadness and concern.

3 How does the unit emphasize knowing your audience before designing any visualization?

Importance of Knowing Your Audience Before Designing a Visualization

Before creating any data visualization, it is important to understand the audience because different people need different types of information and presentation styles.

The unit emphasizes that a visualization should be designed according to:

- Who will view it
- Their level of knowledge
- Their purpose for using the data
- The type of decisions they need to make

For example:

- Managers may prefer simple dashboards, summaries, and key trends.
- Analysts may need detailed graphs, comparisons, and exact values.
- General public may understand better through colorful charts, icons, and simple explanations.

If the audience is not considered:

- The visualization may become confusing.
- Important information may be misunderstood.
- Users may not be able to make proper decisions.

Key Points to Consider About the Audience

1. Technical knowledge of the audience
2. Purpose of the visualization
3. Level of detail required
4. Preferred chart type
5. Need for emotional or analytical understanding

Example

A sales dashboard for company executives may use simple bar charts and KPIs, while a dashboard for data analysts may include scatter plots, trend lines, and detailed comparisons.

Q-4 When should you use a stacked area chart? Explain its advantages.

Stacked Area Chart

A **stacked area chart** is used to show how different parts contribute to the total value over a period of time. It displays both the overall trend and the contribution of individual categories.

In this chart, the areas are stacked on top of one another, making it easy to see changes in total values as well as changes in each category.

When to Use a Stacked Area Chart

A stacked area chart should be used when:

- You want to show trends over time.
- You want to compare the contribution of multiple categories.
- You want to show how the total value changes over time.
- You want to visualize part-to-whole relationships.

Example

A company wants to show sales of different products over six months:

- Product A
- Product B
- Product C

A stacked area chart can show:

- Total sales each month
- Contribution of each product to total sales

Advantages of Stacked Area Chart

1. **Shows Overall Trend**
    - Helps in understanding how the total value changes over time.
2. **Shows Individual Contribution**
    - Makes it easy to see how much each category contributes to the total.
3. **Useful for Comparing Categories**
    - Allows comparison of different groups in a single chart.
4. **Easy to Understand**
    - Provides a clear visual representation of part-to-whole relationships.
5. **Saves Space**
    - Combines multiple data series into one chart instead of using separate charts.

Limitations

- Difficult to compare categories that are not placed at the bottom.
- Becomes confusing if there are too many categories.

5 What is Sigmoid function?

Sigmoid Function

The **Sigmoid function** is an activation function commonly used in machine learning and neural networks. It converts any input value into an output value between **0 and 1**.

The graph of the sigmoid function is S-shaped, so it is also called the **S-curve function**.

Formula of Sigmoid Function

$$\sigma(x) = \frac{1}{1 + e^{-x}}$$

Where:

- x = Input value
- e = Euler's constant
- $\sigma(x)$ = Output of sigmoid function

Properties of Sigmoid Function

- Output values lie between **0 and 1**
- Large positive values give output close to **1**
- Large negative values give output close to **0**
- Input value 0 gives output:

$$\sigma(0) = 0.5$$

Example

If:

$$x = 0$$

Then:

$$\sigma(0) = \frac{1}{1+e^0} = \frac{1}{2} = 0.5$$

Uses of Sigmoid Function

- Used in neural networks as an activation function
- Used in logistic regression
- Helps convert outputs into probabilities
- Useful in binary classification problems

Advantages

- Smooth and differentiable
- Output can be interpreted as probability
- Easy to understand and implement

Disadvantages

- Suffers from vanishing gradient problem
- Output is not centered around zero
- Slower than some modern activation functions like ReLU

6 Explain the Big Three with respect to data visualization.

Big Three with Respect to Data Visualization

The “Big Three” in data visualization refers to the three most important things that should be considered before creating any chart or graph:

1. Audience
2. Purpose
3. Data

These three factors help in designing an effective and meaningful visualization.

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1. Audience

The audience means the people who will view the visualization.

Different audiences require different styles of presentation:

- Executives may want simple dashboards and summaries.
- Analysts may need detailed charts and exact values.

- General public may prefer simple graphs with labels and colors.

Understanding the audience helps in selecting the right chart type, level of detail, and complexity.

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2. Purpose

Purpose means the reason for creating the visualization.

Before designing a chart, we should know:

- What message should be communicated?
- What decision should be supported?
- What trend or pattern should be highlighted?

For example:

- To compare values → Use a bar chart
- To show trends over time → Use a line chart
- To show proportions → Use a pie chart

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3. Data

Data is the information that will be visualized.

The type, size, and quality of data affect the choice of visualization.

- Numerical data may use line charts or histograms.
- Categorical data may use bar charts or pie charts.
- Large datasets may need dashboards or interactive charts.

Correct and clean data is necessary for accurate visualization.

Importance of the Big Three

- Helps create clear and useful visualizations
- Improves understanding of the data
- Avoids confusion and misleading charts
- Makes it easier for viewers to take decisions

7 When should you use an overlapping area chart? Provide reasoning and key features.

Overlapping Area Chart

An **overlapping area chart** is used when two or more data series need to be compared over the same time period or category range. In this chart, the areas are drawn on top of each other with some transparency so that the overlapping regions can be seen clearly.

When to Use an Overlapping Area Chart

An overlapping area chart should be used when:

- You want to compare multiple trends over time.
- You want to see similarities and differences between datasets.
- You want to identify where one category is larger or smaller than another.
- The number of data series is small, usually two or three.

Example

Suppose a company wants to compare:

- Sales of Product A
- Sales of Product B

across different months.

An overlapping area chart can show:

- Which product had higher sales
- Where the sales values overlap
- How the trends change over time

Key Features of Overlapping Area Chart

1. **Displays Multiple Data Series**
    - Helps compare two or more datasets on the same chart.
2. **Uses Transparency**
    - Overlapping regions remain visible because colors are partially transparent.
3. **Shows Trends Over Time**
    - Useful for identifying increases, decreases, and fluctuations.
4. **Highlights Differences**
    - Makes it easier to see which category dominates at different points.
5. **Visual Appeal**
    - More attractive than simple line charts when comparing areas.

Advantages

- Easy comparison of multiple categories
- Highlights overlap and variation
- Good for trend analysis
- Makes patterns easier to observe

Limitations

- Can become confusing if too many categories are added
- Overlapping colors may make values difficult to read
- Exact values are harder to identify compared to line charts

8 Write a note on principles of Data Visualization Design.

Principles of Data Visualization Design

Data visualization design principles are guidelines that help in creating charts, graphs, and dashboards that are easy to understand, attractive, and effective.

Good visualization should communicate information clearly without confusing the audience.

Main Principles of Data Visualization Design

1. **Clarity**
    - The visualization should be simple and easy to understand.
    - Avoid unnecessary decorations, extra colors, or too much text.
2. **Accuracy**
    - Data should be represented correctly.
    - The chart should not mislead the viewer by changing scales or proportions.
3. **Simplicity**
    - Use only the necessary elements.
    - A simple chart is easier to understand than a very complex one.
4. **Consistency**
    - Use the same colors, fonts, symbols, and styles throughout the visualization.
    - Consistency makes the chart look professional and easier to read.
5. **Proper Chart Selection**
    - Choose the right type of chart based on the data:
        - Bar chart for comparison
        - Line chart for trends
        - Pie chart for proportions
        - Scatter plot for relationships
6. **Use of Colors**
    - Colors should highlight important information.
    - Too many colors can confuse the viewer.
    - Similar colors should represent similar categories.
7. **Highlight Important Data**

    - Important points or trends should be emphasized using bold colors, labels, or annotations.
8. **Know the Audience**
    - The chart should be designed according to the knowledge level and needs of the audience.
9. **Avoid Information Overload**
    - Too much data in one chart makes it difficult to understand.
    - Keep the visualization clean and focused.
10. **Readable Labels and Titles**

- Use clear titles, axis labels, legends, and captions.
- This helps the audience understand the meaning of the chart quickly.

Importance of Good Visualization Design

- Improves understanding of data
- Helps in decision-making
- Saves time
- Makes trends and patterns easier to identify
- Avoids confusion and misinterpretation

9 What is an area chart? Describe its two main types with use cases.

Area Chart

An **area chart** is a type of graph that displays quantitative data over time. It is similar to a line chart, but the area below the line is filled with color or shading. This makes it easier to visualize the magnitude of values and compare changes over time.

Area charts are mainly used to show:

- Trends over time
- Magnitude of change
- Part-to-whole relationships

Two Main Types of Area Chart

1. Stacked Area Chart

In a stacked area chart, multiple data series are placed one above another. The chart shows both the total value and the contribution of each category over time.

Features

- Displays cumulative totals
- Shows how each category contributes to the total
- Useful for comparing multiple categories together

Use Cases

- Sales of different products over months
- Population growth of different regions
- Revenue contribution by departments

Example

A company wants to compare sales of Product A, Product B, and Product C over a year. A stacked area chart shows total sales and contribution of each product.

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2. Overlapping Area Chart

In an overlapping area chart, two or more area charts are drawn on the same graph with transparency. This helps compare the trends of different categories directly.

Features

- Shows comparison between multiple datasets
- Highlights overlaps and differences
- Best when the number of categories is small

Use Cases

- Comparing sales of two products
- Comparing rainfall in two cities
- Comparing profits of two companies

Example

A company compares monthly sales of Product A and Product B. The overlapping area chart shows which product performed better in different months.

Advantages of Area Charts

- Easy to understand
- Good for trend analysis
- Shows volume and change over time
- Attractive visual presentation

Limitations

- Can become confusing with too many categories
- Difficult to read exact values
- Overlapping areas may hide smaller values

10 Explain Logistic Regression.

Logistic Regression

Logistic Regression is a supervised machine learning algorithm used for **classification problems**. It predicts the probability that a given input belongs to a particular class.

Unlike linear regression, which predicts continuous values, logistic regression predicts categorical outcomes such as:

- Yes / No
- True / False
- Spam / Not Spam
- Disease / No Disease

Definition

Logistic Regression is a statistical method used to model the relationship between input features and a binary output variable.

Output of Logistic Regression

The output of logistic regression lies between **0 and 1** and represents probability.

- If output > 0.5 → Class 1 (Yes)
- If output < 0.5 → Class 0 (No)

Logistic Function (Sigmoid Function)

Logistic Regression uses the sigmoid function:

$$P(Y=1) = \frac{1}{1+e^{-z}}$$

Where:

$$z = b_0 + b_1x_1 + b_2x_2 + \cdots + b_nx_n$$

- b_0 = Intercept
- $b_1, b_2, \ldots, b_n$ = Coefficients
- $x_1, x_2, \ldots, x_n$ = Input features

The sigmoid function converts the output into a value between 0 and 1.

Working of Logistic Regression

1. Input features are provided to the model.
2. A weighted sum of features is calculated.
3. The sigmoid function converts the result into probability.
4. Based on a threshold value (usually 0.5), the final class is predicted.

Types of Logistic Regression

1. **Binary Logistic Regression**
   - Two possible classes
   - Example: Pass/Fail
2. **Multinomial Logistic Regression**
   - More than two classes
   - Example: Classifying fruits as Apple, Mango, or Orange
3. **Ordinal Logistic Regression**
   - Classes have an order
   - Example: Low, Medium, High

Applications

- Email spam detection
- Disease prediction
- Credit risk analysis
- Customer churn prediction
- Sentiment analysis

Advantages

- Simple and easy to implement
- Works well for binary classification
- Gives probability-based output
- Requires less computational power

Limitations

- Works best only for linear relationships
- Not suitable for very complex data
- Sensitive to outliers

11 How do you add context in a data visualization? Explain the three main methods.

Adding Context in Data Visualization

Context in data visualization means providing extra information that helps the audience understand the meaning of the data more clearly. Without context, a chart may be confusing or misleading.

Adding context helps viewers understand:

- Why the data is important
- What trends or patterns mean
- How to interpret the values correctly

Three Main Methods to Add Context

1. Titles and Labels

Titles, subtitles, axis labels, legends, and captions help explain what the chart represents.

Examples

- Chart title: “Monthly Sales Growth in 2025”
- X-axis label: Months
- Y-axis label: Sales Amount

Importance

- Makes the chart easier to understand
- Clearly tells the audience what data is being shown
- Reduces confusion

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2. Annotations and Notes

Annotations are small text notes added directly to the chart to highlight important points, sudden changes, or unusual events.

Examples

- Marking a sudden drop in sales due to a strike
- Highlighting a peak in website traffic during a festival
- Showing when a new product was launched

Importance

- Helps explain why something happened
- Draws attention to important events
- Makes the visualization more informative

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3. Reference Lines and Benchmarks

Reference lines are lines added to the chart to compare the data with a target, average, or standard value.

Examples

- Average sales line
- Target revenue line
- Minimum passing marks line

Importance

- Helps compare actual performance with expected performance
- Makes it easier to identify above-average or below-average values
- Helps in decision-making

12 What are Statistical Plots?

Statistical Plots

Statistical plots are graphical methods used to represent, summarize, and analyze data. They help in understanding the distribution, relationship, variation, and pattern of data in an easy and visual form.

Instead of studying a large table of numbers, statistical plots make it easier to identify trends, compare values, and draw conclusions.

Definition

A statistical plot is a visual representation of data used to study the behavior and characteristics of a dataset.

Importance of Statistical Plots

- Makes data easier to understand
- Helps identify trends and patterns
- Useful for comparing different datasets
- Helps in detecting outliers and unusual values
- Supports decision-making and analysis
- Useful in statistics, business, research, and machine learning

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Common Types of Statistical Plots

1. Histogram

A histogram is used to show the frequency distribution of continuous numerical data.

- Data is divided into intervals called bins.
- The height of each bar shows the frequency of values in that interval.
- Useful for understanding data distribution.

Example: Distribution of marks scored by students.

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2. Box Plot

A box plot is used to display:

- Median
- Quartiles
- Minimum and maximum values
- Outliers

It helps in understanding the spread and symmetry of data.

Example: Comparing salary distributions of employees.

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3. Scatter Plot

A scatter plot is used to show the relationship between two variables.

- Each point represents one observation.
- Helps identify positive, negative, or no correlation.

Example: Relationship between study hours and exam marks.

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4. Bar Chart

A bar chart is used to compare values across different categories.

- Categories are shown on one axis.
- Values are shown on the other axis.

Example: Comparing sales of different products.

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5. Line Chart

A line chart is used to show changes or trends over time.

- Data points are connected using lines.
- Useful for time series analysis.

Example: Monthly sales of a company.

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6. Pie Chart

A pie chart is used to show the proportion or percentage of each category in a whole.

- The whole circle represents 100%.
- Each slice represents a category.

Example: Market share of different companies.

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7. Density Plot

A density plot is similar to a histogram but uses a smooth curve to show the distribution of data.

- Useful for continuous data
- Helps understand the shape of the distribution

Applications of Statistical Plots

- Business analysis
- Scientific research
- Medical studies
- Financial forecasting
- Machine learning and data science
- Educational performance analysis

Advantages of Statistical Plots

- Easy to understand and interpret
- Saves time in analysis
- Makes large datasets simpler
- Helps in quick comparison
- Visually attractive

Limitations of Statistical Plots

- Wrong plot selection may mislead the audience
- Too much data can make plots confusing
- Some plots are difficult for non-technical users to understand

13 Compare the two main design styles: calculating/exacting response vs. strong emotional response.

Comparison between Calculating/Exacting Response and Strong Emotional Response in Data Visualization

Basis	Calculating / Exacting Response	Strong Emotional Response
Meaning	Focuses on logical thinking, analysis, and exact values	Focuses on emotions, feelings, and personal impact
Objective	Help viewers understand data accurately	Make viewers emotionally connected to the data
Main Focus	Numbers, trends, comparisons, percentages	Human stories, colors, icons, images
Viewer Reaction	“What does this data mean?”	“How does this data make me feel?”
Use of Data	Detailed and precise data	Simplified and impactful data
Visualization Style	Tables, bar charts, line charts, scatter plots	Infographics, pictograms, photos, highlighted colors

Best Used For	Business reports, finance, research, scientific analysis	Social campaigns, health awareness, environment, charity
Decision Making	Supports logical and rational decisions	Supports emotional awareness and action
Color Usage	Neutral and professional colors	Bright, bold, or dramatic colors
Detail Level	High level of detail	Less detail, more visual impact

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Calculating / Exacting Response

This style is used when the goal is to help viewers carefully analyze the data. It focuses on exact numbers, percentages, trends, and comparisons.

Features

- Accurate and detailed
- Helps in logical decision-making
- Uses charts like line graphs, bar charts, and tables
- Suitable for technical audiences

Example

A financial dashboard showing yearly profit, expenses, and sales growth.

Strong Emotional Response

This style is used when the goal is to create feelings such as sadness, fear, concern, excitement, or motivation.

Features

- Uses storytelling and visual impact
- May include colors, images, icons, and annotations
- Helps people connect emotionally with the message
- Suitable for awareness campaigns and social issues

Example

A chart showing accident deaths using red colors and human-shaped icons to create emotional impact.

14 Explain K-Means.

K-Means Clustering

K-Means is an unsupervised machine learning algorithm used to divide data into a fixed number of groups or clusters. It groups similar data points together and separates dissimilar data points into different clusters.

The letter **K** represents the number of clusters to be formed.

Definition

K-Means is a clustering algorithm that partitions a dataset into K groups, where each data point belongs to the cluster with the nearest mean value (centroid).

Main Idea

- Data points that are similar are grouped together.
- Each cluster has a center point called a **centroid**.
- The algorithm repeatedly updates the centroids until the clusters become stable.

Steps of K-Means Algorithm

1. Choose the number of clusters, K.
2. Select K initial centroids randomly.
3. Assign each data point to the nearest centroid.
4. Recalculate the centroid of each cluster.
5. Repeat the process until no significant change occurs in the clusters.

Example

Suppose a company wants to divide customers into 3 groups based on:

- Age
- Income
- Spending habits

Then:

K=3K = 3K=3

The algorithm will create 3 customer groups with similar characteristics.

Applications of K-Means

- Customer segmentation
- Image segmentation
- Document classification
- Market analysis
- Recommendation systems

Advantages

- Simple and easy to understand
- Fast for large datasets
- Works well when clusters are clearly separated

Limitations

- Need to choose K in advance
- Sensitive to outliers
- Does not work well for irregular-shaped clusters
- Different initial centroids may give different results

Important Terms

Cluster

A group of similar data points.

Centroid

The center point of a cluster.

K

The number of clusters to be formed.

Q-15 Explain kNN with one math example.

k-Nearest Neighbors (kNN)

k-Nearest Neighbors (kNN) is a supervised machine learning algorithm used for classification and regression. It classifies a new data point based on the majority class of its nearest neighbors.

The value of **K** represents the number of nearest neighbors considered for prediction.

Working of kNN

1. Choose the value of K.
2. Calculate the distance between the new data point and all existing data points.
3. Select the K nearest neighbors.
4. Find the majority class among those neighbors.
5. Assign that class to the new data point.

Distance Formula

The most common distance measure used in kNN is Euclidean Distance:

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Example

Suppose we have the following classified points:

Point	Coordinates	Class
A	(1,1)	Red
B	(2,2)	Red
C	(4,4)	Blue
D	(5,5)	Blue

We want to classify a new point:

$$P = (3, 3)$$

Let:

$$K = 3$$

Now calculate the distance of P from each point.

Distance from A(1,1)

$$d = \sqrt{(3-1)^2 + (3-1)^2}$$
$$= \sqrt{4+4}$$
$$= \sqrt{8} \approx 2.83$$

Distance from B(2,2)

$$d = \sqrt{(3-2)^2 + (3-2)^2}$$
$$= \sqrt{1+1}$$
$$= \sqrt{2} \approx 1.41$$

Distance from C(4,4)

$$d = \sqrt{(3-4)^2 + (3-4)^2}$$
$$= \sqrt{1+1}$$
$$= \sqrt{2} \approx 1.41$$

Distance from D(5,5)

$$d = \sqrt{(3-5)^2 + (3-5)^2}$$
$$= \sqrt{4+4}$$
$$= \sqrt{8} \approx 2.83$$

Nearest 3 Neighbors

- B → Red
- C → Blue
- A → Red

Among the 3 nearest neighbors:

- Red = 2
- Blue = 1

So the new point (3,3)(3,3)(3,3) will be classified as:

Red

Advantages of kNN

- Simple and easy to understand
- No training phase required
- Works well for small datasets

Limitations

- Slow for large datasets
- Sensitive to outliers
- Choice of K affects accuracy

16 Discuss how to define the purpose of a data visualization after brainstorming.

Defining the Purpose of a Data Visualization after Brainstorming

After brainstorming, the next important step is to clearly define the purpose of the data visualization. This means deciding what message the visualization should communicate and what action or understanding is expected from the audience.

Without a clear purpose, the visualization may become confusing or ineffective.

Steps to Define the Purpose

1. Identify the Main Goal

Determine what you want the audience to learn from the data.

Possible goals include:

- Showing a trend
- Comparing categories
- Explaining relationships
- Identifying outliers
- Showing part-to-whole contribution

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2. Decide the Key Message

Choose the most important point that the audience should remember after viewing the visualization.

For example:

- Sales increased over time
- Product A performed better than Product B
- There is a strong relationship between two variables

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3. Understand the Audience

The purpose depends on who will use the visualization.

- Managers may need a summary for decision-making.
- Analysts may require detailed patterns and exact values.
- Customers may need simple and attractive visuals.

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4. Select the Right Type of Visualization

The purpose helps in selecting the appropriate chart.

- Comparison → Bar Chart
- Trend over time → Line Chart
- Distribution → Histogram
- Relationship → Scatter Plot
- Part-to-whole → Pie Chart

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5. Focus on Action or Decision

A good visualization should help the audience:

- Make a decision
- Understand a problem
- Identify an opportunity
- Take action based on the data

Example

Suppose a company wants to visualize monthly sales data.

After brainstorming, the purpose may be:

- To show how sales changed over time
- To identify the month with highest sales
- To help managers plan future strategies

In this case, a line chart may be the best choice.

Importance of Defining Purpose

- Makes the visualization clear and focused
- Helps choose the correct chart type
- Avoids unnecessary information
- Improves communication and understanding

17 Outline the step-by-step process for designing to meet the needs of your target audience.

Step-by-Step Process for Designing to Meet the Needs of the Target Audience

To create an effective data visualization, it is important to design it according to the needs of the target audience. Different users require different levels of detail, chart styles, and information.

Step 1: Identify the Target Audience

First, determine who will use or view the visualization.

Examples:

- Managers
- Analysts
- Customers
- Students
- General public

Different audiences have different levels of knowledge and expectations.

Step 2: Understand Audience Needs

Find out what information the audience wants and why they need it.

Questions to consider:

- What decisions will they make using the data?
- Do they need detailed or summarized information?
- Are they technical or non-technical users?

Step 3: Define the Main Objective

Decide the main purpose of the visualization.

Examples:

- Show trends
- Compare values
- Identify patterns
- Detect outliers

- Explain relationships

Step 4: Choose the Right Data

Select only the most relevant data needed for the audience.

Too much information can confuse users, so unnecessary data should be removed.

Step 5: Select the Appropriate Chart Type

Different chart types are suitable for different purposes:

- Bar Chart → Comparison
- Line Chart → Trend over time
- Pie Chart → Part-to-whole relationship
- Scatter Plot → Relationship between variables
- Histogram → Distribution of data

Step 6: Use Suitable Colors and Design

Colors, fonts, labels, and layout should match the audience.

- Use simple colors for business reports
- Use attractive visuals for general public
- Highlight important information using bold colors

Step 7: Add Labels, Titles, and Context

A good visualization should include:

- Clear title
- Axis labels
- Legends
- Notes or annotations

This helps the audience understand the data more easily.

Step 8: Test and Improve the Visualization

Before finalizing, test the visualization with users.

Check:

- Is it easy to understand?
- Is any information missing?
- Is the chart confusing?

Make improvements based on feedback.

Importance

- Makes visualization more useful
- Improves understanding
- Helps audience make better decisions
- Increases clarity and effectiveness

18 Write a short note on Linear Regression with example.

Linear Regression

Linear Regression is a supervised machine learning algorithm used to predict a continuous numerical value based on the relationship between one dependent variable and one or more independent variables.

It is called “linear” because it assumes that the relationship between variables can be represented by a straight line.

Definition

Linear Regression is a statistical method used to find the relationship between input variables and an output variable by fitting a straight line to the data.

Equation of Linear Regression

Y=a+bXY = a + bXY=a+bX

Where:

- YYY = Dependent variable
- XXX = Independent variable
- aaa = Intercept
- bbb = Slope of the line

The equation shows how much YYY changes when XXX changes.

Working of Linear Regression

1. Collect the input and output data.
2. Find the best-fit straight line.
3. Use the line equation to predict new values.

Example

Suppose a company wants to predict sales based on advertising expenses.

Advertising Expense (X)	Sales (Y)
1	2
2	4
3	6
4	8

The relationship is:

Y=2XY = 2XY=2X

If advertising expense is 5:

Y=2(5)=10Y = 2(5) = 10Y=2(5)=10

So, predicted sales = 10.

Applications

- House price prediction
- Sales forecasting
- Weather forecasting
- Stock market analysis
- Demand prediction

Advantages

- Simple and easy to understand
- Useful for predicting continuous values
- Fast and efficient

Limitations

- Works only for linear relationships
- Sensitive to outliers
- Cannot handle complex patterns well

19 Write a note on Chart Graphics used in data visualization (e.g. Bar Chart, Pie Chart etc.)

Chart Graphics Used in Data Visualization

Chart graphics are visual tools used to represent data in an easy and understandable form. Different types of charts are used depending on the type of data and the purpose of analysis.

Charts help in comparing values, identifying trends, understanding relationships, and making decisions.

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1. Bar Chart

A bar chart uses rectangular bars to compare values among different categories.

Features

- Length of the bar represents the value.
- Bars can be horizontal or vertical.
- Useful for comparing different groups.

Example

Comparing sales of different products.

---

2. Pie Chart

A pie chart is a circular chart divided into slices. Each slice represents a percentage or proportion of the whole.

Features

- Whole circle represents 100%.
- Useful for showing part-to-whole relationships.

Example

Market share of different companies.

---

3. Line Chart

A line chart uses points connected by lines to show changes over time.

Features

- Useful for trend analysis.
- Helps show increase or decrease over time.

Example

Monthly sales of a company.

---

4. Histogram

A histogram is used to show the frequency distribution of numerical data.

Features

- Data is divided into intervals called bins.
- Bars touch each other because data is continuous.

Example

Distribution of student marks.

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5. Scatter Plot

A scatter plot shows the relationship between two variables.

Features

- Each point represents one observation.
- Helps identify positive, negative, or no correlation.

Example

Relationship between study hours and exam marks.

---

6. Area Chart

An area chart is similar to a line chart, but the area under the line is filled with color.

Features

- Shows trends and magnitude over time.
- Useful for comparing totals.

Example

Monthly revenue growth.

---

7. Box Plot

A box plot is used to show:

- Median
- Quartiles
- Outliers
- Spread of data

Example

Comparing salary distributions.

---

8. Bubble Chart

A bubble chart is similar to a scatter plot, but the size of the bubble represents an additional variable.

Example

Comparing countries by population, GDP, and area.

---

9. Heat Map

A heat map uses colors to represent values in a matrix or table.

Example

Attendance record of students.

Importance of Chart Graphics

- Makes data easy to understand
- Helps in quick comparison
- Shows trends and relationships clearly
- Useful in business, research, and machine learning

20 What ethical responsibility does a data scientist have while creating visualizations?

Ethical Responsibility of a Data Scientist while Creating Visualizations

A data scientist has an important ethical responsibility to create visualizations that are truthful, fair, accurate, and easy to understand. Misleading or manipulated visualizations can result in wrong decisions and misunderstanding.

Main Ethical Responsibilities

1. Present Data Honestly

The data should be shown exactly as it is without changing or hiding important information.

- Do not exaggerate values
- Do not remove important data points
- Do not manipulate scales to create false impressions

---

2. Avoid Misleading Visuals

Charts should not be designed in a way that confuses or misleads the audience.

Examples of misleading practices:

- Using incorrect axis scales
- Using 3D charts that distort values
- Using inappropriate chart types

---

3. Use Correct and Complete Data

A data scientist should ensure that:

- Data is accurate

- Data is complete
- Data is collected from reliable sources

Incomplete or incorrect data can lead to false conclusions.

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4. Respect Privacy and Confidentiality

Sensitive personal information should not be exposed in visualizations.

Examples:

- Patient medical records
- Personal contact details
- Financial information

Private information should be hidden or anonymized.

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5. Avoid Bias

The visualization should be neutral and unbiased.

- Do not highlight only the information that supports one viewpoint
- Show both positive and negative aspects of the data
- Avoid using colors or labels that unfairly influence the audience

---

6. Make Visualizations Easy to Understand

The audience should be able to understand the chart without confusion.

- Use clear labels and titles
- Use proper legends
- Avoid unnecessary complexity

Importance of Ethics in Visualization

- Builds trust
- Supports correct decision-making
- Prevents misinformation
- Protects privacy
- Encourages responsible use of data

21 Write Python code using Scikit-learn and Matplotlib to demonstrate the end-to-end process of model fitting, prediction, and data visualization for the specified algorithms. Linear Regression Model kNN Classifier (take k=3)

Linear Regression Model using Scikit-learn and Matplotlib

```
import numpy as np
import matplotlib.pyplot as plt
from sklearn.linear_model import LinearRegression

# Sample data
X = np.array([1, 2, 3, 4, 5]).reshape(-1, 1)
y = np.array([2, 4, 6, 8, 10])

# Create and train the model
model = LinearRegression()
model.fit(X, y)

# Prediction
y_pred = model.predict(X)

# Predict for a new value
new_value = np.array([[6]])
predicted_value = model.predict(new_value)
print("Predicted value for X = 6:", predicted_value[0])

# Visualization
plt.scatter(X, y, color='blue', label='Actual Data')
plt.plot(X, y_pred, color='red', label='Regression Line')
plt.xlabel('X Values')
plt.ylabel('Y Values')
plt.title('Linear Regression Model')
plt.legend()
plt.show()
```

kNN Classifier (k = 3) using Scikit-learn and Matplotlib

```python
import numpy as np
import matplotlib.pyplot as plt
from sklearn.neighbors import KNeighborsClassifier

# Sample training data
X = np.array([
    [1, 1],
    [2, 2],
    [3, 3],
    [6, 6],
    [7, 7],
    [8, 8]
])

y = np.array([0, 0, 0, 1, 1, 1])

# Create kNN model with k = 3
model = KNeighborsClassifier(n_neighbors=3)

# Train the model
model.fit(X, y)

# Predict class for a new point
new_point = np.array([[4, 4]])
prediction = model.predict(new_point)
print("Predicted Class for point [4,4]:", prediction[0])

# Visualization
for i in range(len(X)):
    if y[i] == 0:
        plt.scatter(X[i][0], X[i][1], color='blue')
    else:
        plt.scatter(X[i][0], X[i][1], color='green')

# Plot new point
plt.scatter(new_point[0][0], new_point[0][1], color='red', marker='x', s=100, label='New Point')

plt.xlabel('Feature 1')
plt.ylabel('Feature 2')
plt.title('kNN Classifier (k=3)')
plt.legend()
plt.show()
```

Explanation

Linear Regression

1. Import dataset
2. Train the Linear Regression model
3. Predict output values
4. Plot actual points and regression line

kNN Classifier

1. Create training data with two classes
2. Train the kNN model with k = 3
3. Predict the class of a new point
4. Visualize training data and predicted point